

Adaptive Sampling Based Digital Communication

Abstract :

This project presents an efficient end-to-end communication system designed to transmit analog signals with high integrity and minimal data redundancy. Conventional uniform sampling often leads to excessive data transmission in regions where the signal remains relatively constant. To address this, we implement an Adaptive Sampling algorithm that dynamically adjusts the sampling rate based on the signal's slope.

The system utilizes two Arduino Uno microcontrollers connected via a UART digital link. The transmitter digitizes and filters the input, while the receiver decodes and reconstructs the physical analog voltage using a passive RC low-pass filter.

Objectives:

- **Bandwidth Optimization:** To reduce the total number of transmitted samples by implementing event-based adaptive sampling.
- **Noise Immunity:** To establish a robust digital UART communication link that eliminates the electrical interference common in long-distance analog lines.
- **Physical Signal Reconstruction:** To successfully recreate a smooth, physical analog voltage from digital data using PWM and hardware filtering.
- **Real-Time Data Visualization:** To provide immediate monitoring of system performance through an I2C-based LCD interface.

Components Used :

1. Input Stage

- 10kOhm Potentiometer: Serves as the analog message source, allowing for manual variation of the input voltage (0V to 5V).
- 100nF Capacitor: Acts as a noise filter at the input stage to stabilize the readings before they enter the microcontroller.

2. Transmitter (Arduino 1)

- Arduino UNO: Executes the adaptive sampling algorithm by comparing the current input (A0) against a slope threshold.

- UART TX Port: Converts the processed analog values into digital packets for transmission over the communication channel.

3. Communication Channel

- UART Link (TX to RX): A 3-wire digital connection (TX, RX, and Common Ground) that transfers data packets at 9600 baud.

4. Receiver (Arduino 2)

- Arduino UNO: Receives digital packets via its RX pin and decodes the numerical values for reconstruction.
- 16x2 I2C LCD: Displays the received digital count and the corresponding reconstructed voltage in real-time.

5. Output Reconstruction Stage

- PWM Output: The receiver generates a Pulse Width Modulated signal proportional to the received data.
- RC Filter (1kOhm resistor + 1uF Capacitor): A hardware-based low-pass filter that smooths the digital pulses into a steady analog DC output

Block Diagram :

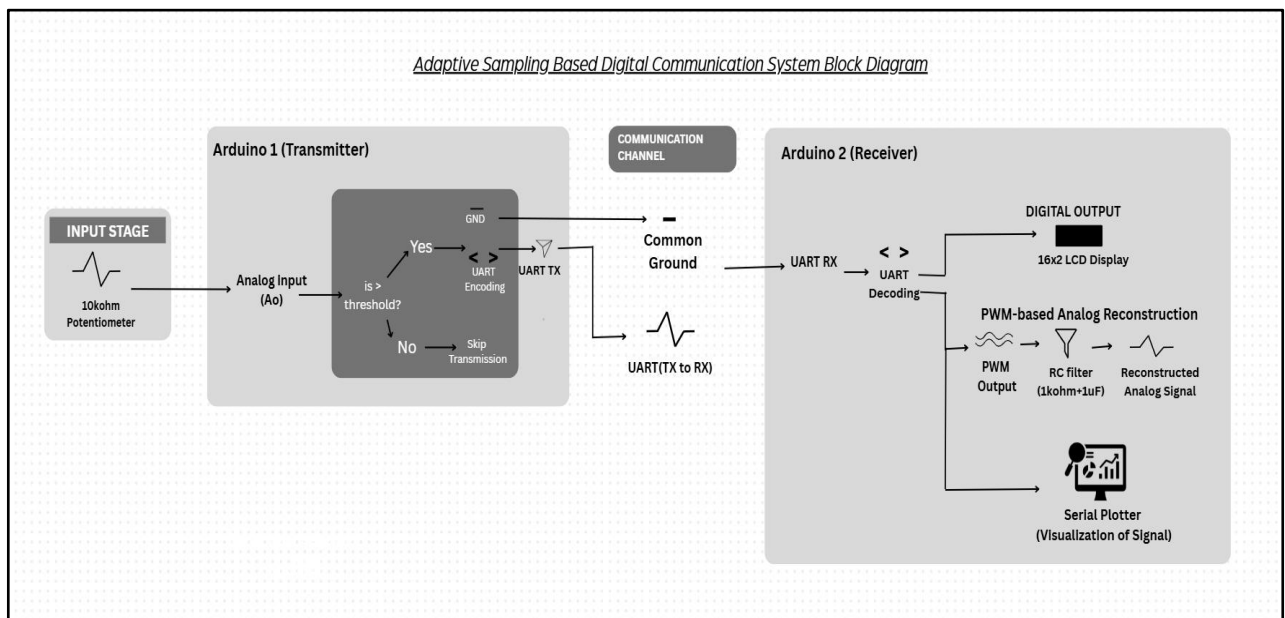
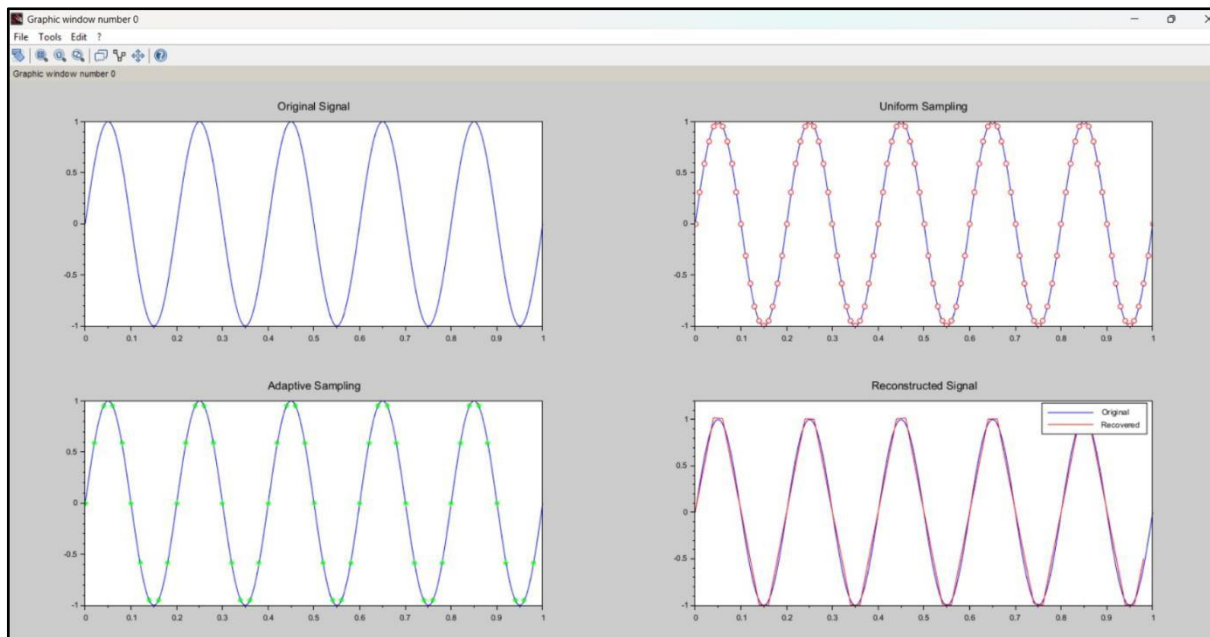


Fig.1:Functional Block Diagram of the Adaptive Sampling-Based Communication System

Fig.1 illustrates the the end-to-end signal flow from analog input and adaptive sampling at the transmitter to digital UART transmission and hardware-based RC filter reconstruction at the receiver.

Simulation Results and Analysis:



(a)

```
Enter Signal Frequency (Hz) : 5
Enter Slope Threshold (e.g., 0.02) : 0.15
Enter Number of Quantization Levels (e.g., 8) : 8
Enter Uniform Sampling Interval (e.g., 0.01) : 0.01

"----- SAMPLE COMPARISON -----"
"Uniform Samples = 101"
"Adaptive Samples = 50"
"Sample Reduction (%) = 50.49505"
```

(b)

Fig.2: Scilabs Simulation (a) Graphical Output (b) Console Results

Fig.2 Scilabs simulation demonstrates the effectiveness of the Adaptive Sampling algorithm in reducing data redundancy. By comparing the uniform and adaptive sampling plots, it is evident that the system skips unnecessary samples during periods of slow signal change, resulting in a 50.5% reduction in total transmitted data (50 samples vs 101). Despite this significant reduction, the Reconstructed Signal graph shows an accurate recovery of the 5Hz sine wave, proving that signal integrity is preserved while optimizing bandwidth.

The diagram illustrates the hardware setup for the Adaptive Sampling Based Digital Communication System. It consists of two Arduino Uno microcontrollers, one configured as a Transmitter and the other as a Receiver.

Transmitter (ARDUINO (Transmitter)):

- Power:** Connected to a 5V supply and GND.
- Input:** A 10K POTENTIOMETER is connected to the 5V and GND rails, with its wiper connected to the A0 pin (14/A0/PC0).
- Communication:** The TX pin (PD1/1) is connected to the RX pin of the Receiver.
- Display:** An LCD 16*02 I2C display is connected to the I2C pins (SCL, SDA, VCC, GND).

Receiver (ARDUINO (Receiver)):

- Power:** Connected to a 5V supply and GND.
- Output:** The TX pin (PD1/1) is connected to a Filter resistor (1k ohm) and a Filter Capacitor (1uF), with an analog output labeled TP1.
- Communication:** The RX pin (PD0/0) is connected to the TX pin of the Transmitter.

Common Ground: Both Arduino boards share a common ground connection.

UART Serial Communication (9600 baud): The two boards are connected via a serial cable for data transfer.

LED (GREEN): A green LED is connected to the RX pin of the Receiver through a 220Ω resistor, likely indicating successful reception.

Fig.3 demonstrates the circuit diagram illustrates a dual- microcontroller architecture where the Transmitter Arduino processes analog input from a potentiometer and communicates with the Receiver Arduino via a 9600 baud UART link. The system features an I2C LCD for data visualization and a hardware-based RC filter for physical signal reconstruction.

The project operates on the principle of Event-Based Adaptive Sampling, which optimizes the communication between two microcontrollers by transmitting data only when significant signal changes occur, rather than at fixed time intervals.

1. Signal Acquisition and Adaptive Logic:

The process begins at the Transmitter (Arduino UNO 1). An analog signal is generated using a 10kOhm potentiometer, which is stabilized by a 100nF noise filter capacitor. The Arduino reads this value through its 10-bit ADC. Instead of continuous transmission, the system applies Adaptive Sampling: it compares the current reading with the previously transmitted value. If the difference is less than the defined Slope Threshold, the sample is discarded. A digital packet is only transmitted if the signal has changed significantly, effectively reducing bandwidth consumption.

2. Digital Data Transmission:

Once the adaptive logic triggers a transmission, the data is sent via the UART (Universal Asynchronous Receiver-Transmitter) protocol at a baud rate of 9600. By converting the analog voltage into a digital format (0–255 or 0–1023) for transmission, the system gains High Noise Immunity. Unlike analog signals, which degrade over distance due to EMI (Electromagnetic Interference), the digital UART link ensures that the received data perfectly matches the transmitted value.

3. Reconstruction and Visualization:

The Receiver (Arduino UNO 2) captures the digital packets. It performs two simultaneous actions:

- Visual Monitoring: It sends the data to a 16x2 I2C LCD, displaying the real-time digital count and voltage for user verification.
- Physical Reconstruction: To recreate the analog signal, the Arduino generates a PWM (Pulse Width Modulation) signal proportional to the received value. This high-frequency square wave is passed through a hardware-based RC Low-Pass Filter (1kOhm resistor and 1u F capacitor). The filter "averages" the PWM pulses, smoothing them back into a steady, physical analog DC voltage that accurately represents the original input.

Breadboard Implementation:

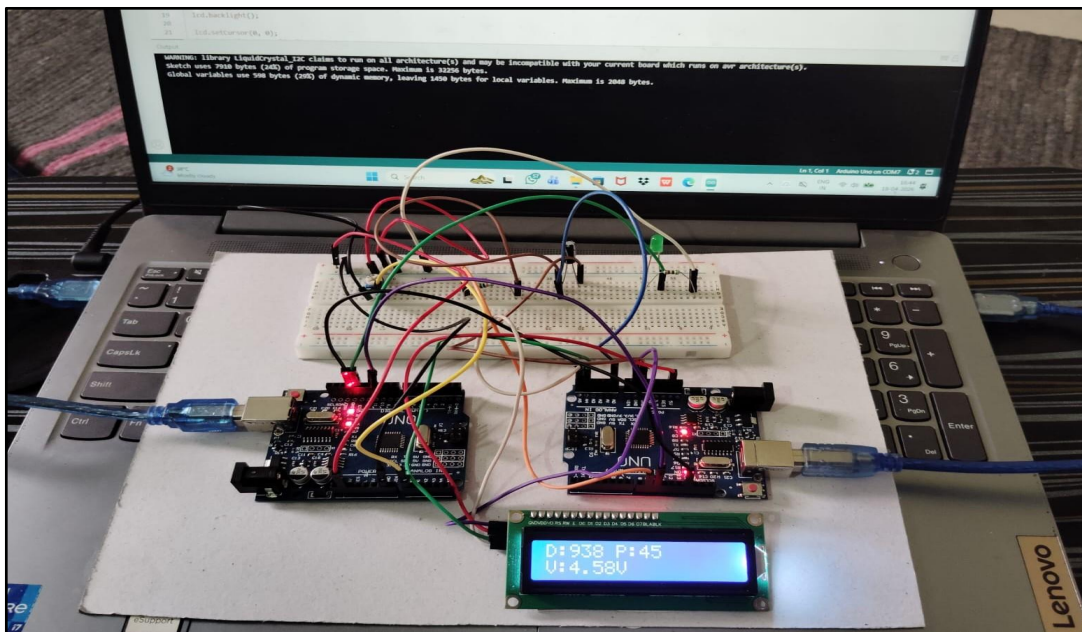


Fig.4:Physical Hardware Implementation on Breadboard

This photograph displays the complete hardware setup, featuring the dual-Arduino transmitter-receiver configuration, I2C LCD interface, and the RC filter circuitry interconnected for real-time validation.

Results :



Fig.5: Serial Plotter Outout demonstrating event-Based Data Transmission

Fig.5 shows Arduino Serial Plotter graph that confirms the real-time execution of the Adaptive Sampling algorithm. The blue line represents the continuous analog input from the potentiometer, while the orange line indicates the fixed slope threshold used for decision-making. The green spikes signify the successful transmission of digital packets, which only occur when the signal change exceeds the threshold, visually demonstrating a significant reduction in data traffic compared to continuous sampling.



Fig.6: serial monitor Log confirming adaptive filtering and Data Reduction

Fig.6 shows Serial Monitor output that provides a real-time log of the Adaptive Sampling logic in action. The text confirms that the system actively evaluates each analog reading against the slope threshold:

- **Packet Sent:** Indicates that a significant change in voltage was detected, triggering a data transmission to the receiver.

- **Packet Skipped:** Demonstrates the core efficiency of the project, where minor signal fluctuations or redundant values (e.g., small shifts between 980 and 984) are filtered out.
- **Result:** This log serves as direct evidence of bandwidth optimization, as the majority of samples are "skipped," ensuring the communication channel is only used when necessary.

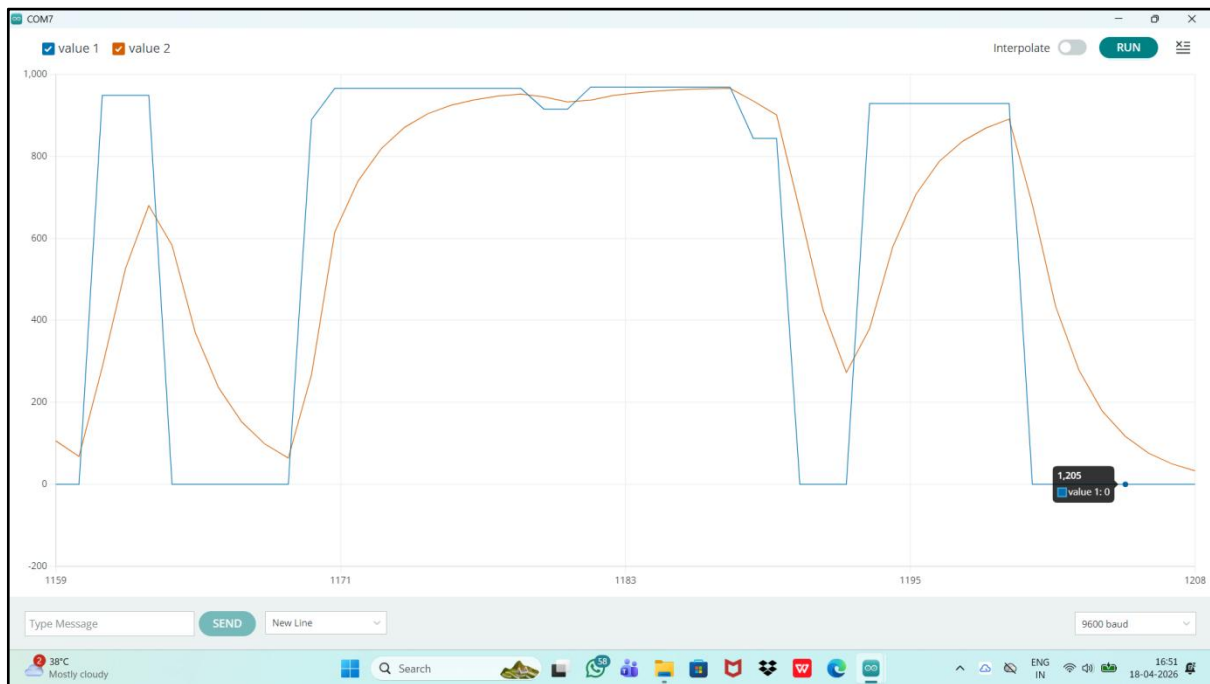


Fig.7:Serial plotter output showing Digital to analog Reconstruction and RC filtering

Fig.7 shows the Receiver Serial Plotter graph that confirms the successful physical reconstruction of the transmitted data. The two waveforms demonstrate the transition from digital reception to a smoothed analog output:

- **Value 1 (Blue Line):** Represents the Raw Digital Data received via UART and converted into a high-frequency PWM signal. The sharp vertical edges show the rapid switching of the digital pulses.
- **Value 2 (Orange Line):** Represents the Reconstructed Analog Signal after passing through the hardware RC Low-Pass Filter. The curve is noticeably smoother, showing how the filter removes the digital switching noise to recreate a steady DC voltage.
- **Result:** The graph validates that despite the adaptive sampling at the transmitter, the receiver is able to effectively track the original signal's behavior and produce a usable, continuous physical voltage.

Observations:

- **Data Efficiency:** The adaptive sampling algorithm successfully reduced data transmission proving that event-based sampling significantly optimizes bandwidth compared to traditional uniform methods.

- **Signal Integrity:** The UART digital link maintained high accuracy and noise immunity, ensuring the received digital values perfectly matched the transmitter's input.
- **Hardware Reconstruction:** The RC low-pass filter effectively smoothed the 490Hz PWM pulses, converting the discrete digital packets back into a continuous, stable physical analog voltage.
- **Real-time Monitoring:** The I2C LCD and Serial Plotter confirmed minimal latency between the physical movement of the potentiometer and the system's response.

Applications :

- **IoT & Wireless Sensor Networks:** Reduces data traffic and saves battery life in remote sensors by transmitting only on significant changes.
- **Industrial Monitoring:** Optimizes bandwidth in factory automation by filtering redundant data from machinery sensors.
- **Environmental Sensing:** Efficiently tracks slow-moving variables like air quality, soil moisture, or water levels.
- **Medical Wearables:** Extends the operating time of devices like heart rate monitors by decreasing the frequency of wireless transmissions.
- **Smart Infrastructure:** Used in bridge or building health monitoring to trigger alerts only when vibration thresholds are exceeded.

Future Scope:

- **Wireless Transmission:** Replacing the UART link with Bluetooth (HC-05), Wi-Fi (ESP32), or LoRa for long-range, cable-free monitoring.
- **Dynamic Thresholding:** Implementing an intelligent algorithm that automatically adjusts the slope sensitivity based on real-time environmental noise.
- **Multi-Node Networking:** Expanding the architecture to a "Star Topology" where a single receiver manages data from multiple transmitter sensor nodes.
- **Cloud & Mobile Integration:** Connecting the system to IoT platforms like ThingSpeak or Blynk for remote data logging and smartphone-based visualization.

Conclusion:

This project successfully validates that an Adaptive Sampling approach can optimize digital communication by achieving a data reduction of over 50% without losing signal integrity. Through the integration of a UART serial link and a hardware RC low-pass filter, the system demonstrated robust noise immunity and accurate analog signal reconstruction. Ultimately, the results prove that event-based sampling is an efficient, scalable solution for reducing bandwidth and power consumption in modern IoT and industrial sensing applications.